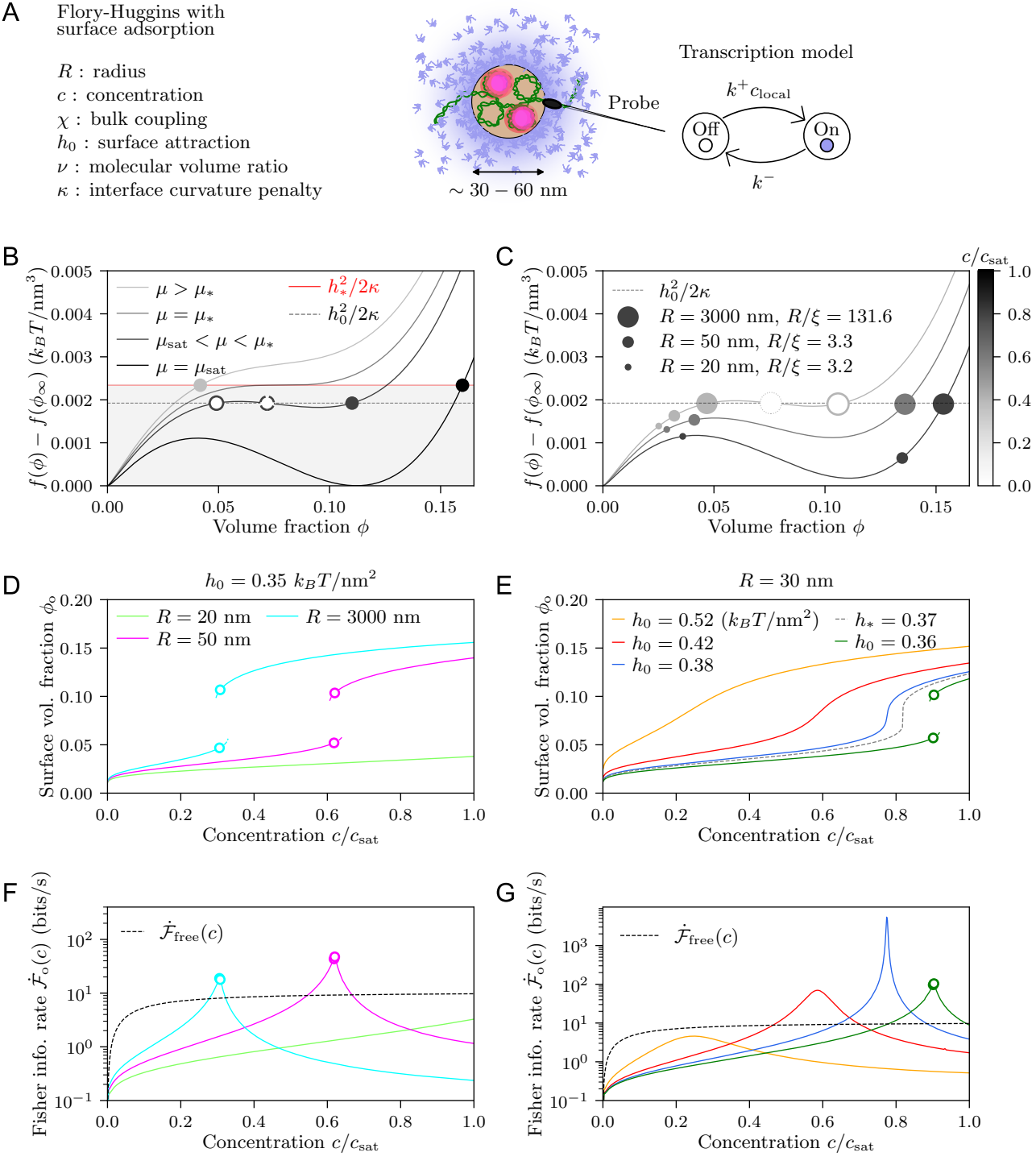

A
Flory-Huggins with surface adsorption
R : radius
c : concentration
χ : bulk coupling
h_0 : surface attraction
ν : molecular volume ratio
κ : interface curvature penalty
$\sim 30-60$ nm
Probe
Transcription model
$k^+ c_{\text{local}}$
Off
On
k^-
B
$f(\phi) - f(\phi_\infty)$ (k_BT/nm^3)
$\mu > \mu_*$
$\mu = \mu_*$
$\mu_{\text{sat}} < \mu < \mu_*$
$\mu = \mu_{\text{sat}}$
$h_*^2/2\kappa$
$h_0^2/2\kappa$
Volume fraction ϕ
C
c/c_{sat}
$h_0^2/2\kappa$
$R = 3000$ nm, $R/\xi = 131.6$
$R = 50$ nm, $R/\xi = 3.3$
$R = 20$ nm, $R/\xi = 3.2$
Volume fraction ϕ
D
$h_0 = 0.35\ k_BT/\text{nm}^2$
Surface vol. fraction ϕ_o
$R = 20$ nm
$R = 3000$ nm
$R = 50$ nm
Concentration c/c_{sat}
E
$R = 30$ nm
Surface vol. fraction ϕ_o
$h_0 = 0.52$ (k_BT/nm^2)
$h_* = 0.37$
$h_0 = 0.42$
$h_0 = 0.36$
$h_0 = 0.38$
Concentration c/c_{sat}
F
Fisher info. rate $\dot{\mathcal{F}}_o(c)$ (bits/s)
$\dot{\mathcal{F}}_{\text{free}}(c)$
Concentration c/c_{sat}
G
Fisher info. rate $\dot{\mathcal{F}}_o(c)$ (bits/s)
$\dot{\mathcal{F}}_{\text{free}}(c)$
Concentration c/c_{sat}